Reg. No. | | | | | | | | | | |

[21MSCH3S03]

THIRD SEMESTER UNIVERSITY EXAMINATION, NOVEMBER 2022

APPLIED CHEMISTRY – I

Time: 03 HOURS

Maximum Marks: 50

SECTION – A

1. **Answer any FOUR of the following questions.** 4×5=20
   - a. Describe carbon nanotubes and their classification.
   - b. Define with examples any two applications of metal oxide quantum dots in detail.
   - c. Write the Scherrer's equation and explain the terminology.
   - d. Describe the applications of EXAFS.
   - e. Explain emulsion polymerization and its advantages.

SECTION – B

2. **Answer any TWO of the following questions.** 2×9=18
   - a. Explain the synthesis procedure and applications of congo red.
   - b. Give the classification of carbonaceous materials. List out the desired characteristics of anode materials for energy storage devices.
   - c. Discuss the theory of homogeneous catalysis. Write any two advantages and limitations.

SECTION – C

3. **Answer the following question.** 12 Marks
   - a. i. Explain in detail condensation polymerization.
   - ii. Write a note on pigments, thinners and surfactants.
   - iii. Differentiate between primary and secondary batteries.

2b. Write a note on ~~elect~~ optical & electrical properties of Colloids

3a iii Define Colloids and Explain Tindall effect in Colloids.

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